Ikhan Choi

June 11, 2025

Contents

1 May 21

1.1 (Simplicial sets)**.** The *simplex category* is the category Δ defined such that objects are totally ordered finite non-empty sets and morphisms are non-decreasing functions. It has a skeleton whose objects are $[n] := \{0, \cdots, n\}$ for $n \geq 0$ with the usual order. In Δ, there are special morphisms as in the following diagram

$$\begin{tikzcd}[0] \arrow[r, shift left=2, "\delta_1^0"] \arrow[r, shift right=2, "\delta_1^1"'] & {[1]} \arrow[l, "\sigma_0^0" description] \arrow[r, shift left=4, "\delta_2^0"] \arrow[r, "\delta_2^1" description] \arrow[r, shift right=4, "\delta_2^2"'] & {[2]} \arrow[l, shift right=2, "\sigma_1^1"'] \arrow[l, shift left=2, "\sigma_1^0"] & \cdots\end{tikzcd}$$

defined such that for every $0 \leq i \leq n$

$$\delta_n^i : [n-1] \to [n] : k \mapsto k + 1_{\geq i}(k), \qquad \sigma_n^i : [n+1] \to [n] : k \mapsto k - 1_{>i}(k),$$

called the *coface maps* and *codegeneracy maps*.

A *simplicial set* is a contravariant functor $X : \Delta^{\mathrm{op}} \to \mathrm{Set}$. We usually write $X_n := X([n])$ to mean the set of *simplices*, and $d_i^n := X(\delta_n^i) : X_n \to X_{n-1}$ and $s_i^n := X(\sigma_n^i) : X_n \to X_{n+1}$ for $0 \leq i \leq n$ are called *face maps* and *degeneracy maps*. It is intuitively something that consists of a sequence X_n of sets that enjoys the combinatorics of face and degeneracy maps. As the image of the skeleton $\{[n]\}$, we have a diagram

$$\begin{tikzcd}\cdots & X_2 \arrow[r, shift left=4, "d_0^2"] \arrow[r, "d_1^2" description] \arrow[r, shift right=4, "d_2^2"'] & X_1 \arrow[l, shift right=2, "s_1^1"'] \arrow[l, shift left=2, "s_0^1"] \arrow[r, shift left=2, "d_0^1"] \arrow[r, shift right=2, "d_1^1"'] & X_0 \arrow[l, "s_0^0" description]\end{tikzcd}$$

A *simplicial map* between simplicial sets is a natural transformation. The category of simplicial sets and simplicial maps is denoted by Set_Δ.

(a) Every morphism of Δ is uniquely decomposed into the composition of codegeneracy maps on the left and coface maps on the right.

1.2 (Simplicial complexes)**.** A *simplicial complex* is a collection X of non-empty finite subsets of a set V closed under non-empty subsets. We denote by X_n for each $n \geq 0$ the set of all functions $[n] \to V$ whose image coincides with an element of X. More categorically, it can be essentially defined as a functor $X : \mathrm{Fin}_{>0}^{\mathrm{op}} \to \mathrm{Set} : [n] \mapsto X_n$ on the category of non-zero finite sets which is concrete in the sense that the canonical map $X_n \to \prod_{\mathrm{Hom}_\Delta([0],[n])} X_0$ is injective for each $n \geq 0$.

If we chose each skeleton of the simplex category and the category of non-empty finite sets, and if we denote the objects of each skeleton as $[n]$ for non-negative integer n, then the forgetful functor $\Delta \to \mathrm{Fin}_{>0}$ gives rise to a linear order on the chosen skeleton of $\mathrm{Fin}_{>0}$, and the composition with the forgetgul functor provides a functor $\mathrm{Cpx}_\Delta \to \mathrm{Set}_\Delta$. We can prove that this is fully faithful.

1.3 (Standard simplices)**.** A *standard simplex* is a simplicial set given by the representable functor in Δ. More precisely, the *standard n-simplex* for $n \geq 0$ is the simplicial set $\Delta^n := \mathrm{Hom}_\Delta(-, [n]) : \Delta^{\mathrm{op}} \to \mathrm{Set}$. For a simplicial set X, the Yoneda lemma implies $X_n = X([n]) = \mathrm{Hom}_{\mathrm{Set}_\Delta}(\Delta^n, X)$. Especially, since $\Delta^n_m = \mathrm{Hom}_\Delta([m],[n]) = \mathrm{Hom}_{\mathrm{Set}_\Delta}(\Delta^m, \Delta^n)$ by the Yoneda lemma, we will identify a monotone function $\alpha : [m] \to [n]$ with the simplicial map $\alpha : \Delta^m \to \Delta^n$.

The set of homotopy classes of simplicial functions $[X, Y]$?

For $0 \leq i \leq n$, *horns* and *boundaries* are simplicial sets Λ_i^n and $\partial\Delta^n$ defined such that

$$\Lambda_i^n([m]) := \{\alpha : [m] \to [n] \mid \alpha([m]) \not\supset [n] \setminus \{i\}\},$$
$$\partial\Delta^n([m]) := \{\alpha : [m] \to [n] \mid \alpha([m]) \neq [n]\}.$$

1.4 (Nerves)**.** We say a simplicial set Y satisfies the *Segal condition* if $Y \to *$ has the unique right lifting property with respect to the horn inclusions $\Lambda_i^n \to \Delta^n$ for all $0 < i < n$.

Let $\mathcal{C}$ be a small category. The *nerve* of $\mathcal{C}$ is a simplicial defined as follows.

The nerve, or its geometric realization, of a small category $\mathcal{C}$ is also called the *classifying space* of $\mathcal{C}$. For example, if we interpret a group G as a category of a single object with invertible morphisms, then its nerve gives rise to its classifying space BG.

(a) The nerve of a small category satisfies the Segal condition.

(b) A simplicial set satisfying the Segal condition is isomoprhic to the nerve of a small category.

(c) The nerve functor is fully faithful.

(d) The nerve functor admits a left adjoint, which is called the *homotopy category functor*.

1.5 (Kan complexes)**.** We say a simplicial set Y satisfies the *Kan filler condition* or Y is a *Kan complex* if $Y \to *$ has the right lifting property with respect to the horn inclusions $\Lambda_i^n \to \Delta^n$ for all $0 \leq i \leq n$.

(a) The singular complex of a topological space is a Kan complex.

(b) A simplicial group is a Kan complex.

(c) The nerve of a groupoid is a Kan complex.

(d) The singular complex functor admits a left adjoint, which is called the *geometric realization*.

$$\mathrm{hCW} \to \mathrm{hCG?} \to \mathrm{Top}[\mathcal{W}^{-1}]$$

The category of simplicial sets is basically the category of presheaves.

2 May 26

- Ivo Dell' Ambrogio, *The unitary symmetric monoidal model category of small C*-categories*
- Emily Riehl, *Homotopical categories: from model categories to* $(\infty, 1)$*-categories*

Theorem (Dell' Ambrogio)**.** $\mathrm{C}^*\mathrm{Cat}$ *is a model category.*

Definition. A *model category* is a bicomplete category with a *model structure*.

Definition (C^*-category)**.** A C^*-category is a category A which is a $\mathbb{C}$-linear category with conjugate-linear involutions $* : A(x, y) \to A(y, x)$ and complete norms $\|\cdot\| : A(x, y) \to [0, \infty)$ on each hom-set $A(x, y)$ such that $\|fg\| \leq \|f\|\|g\|$ and $\|f^*f\| = \|f\|^2$ and $f^*f \geq 0$ are satisfied.

Example. A C^*-algebra is a C^*-category with a single object.

Example. The category Hilb defined such that objects are separable Hilbert spaces and morphisms are bounded linear operators is a C^*-category.

Definition. The category $\mathrm{C}^*\mathrm{Cat}$ is the category of C^*-categories whose morphisms are given by $*$-functors.

Definition. Two C^*-categories A and B are called *unitary equivalent* if there are $*$-functors $F : A \to B$ and $G : B \to A$ such that $GF \simeq \mathrm{id}_A$ and $FG \simeq \mathrm{id}_B$. Here, $GF \simeq \mathrm{id}_A$ means that for each morphism $f : a \to a'$ in A, we have unitaries $u_a : GF(a) \to a$ and $u_{a'} : GF(a') \to a'$ such that $fu_a = u_{a'}GF(f)$. The unitarity of u_a means that $u_a^* u_a = \mathrm{id}_{GF(a)}$ and $u_a u_a^* = \mathrm{id}_a$.

Definition. We define a model structure on $\mathrm{C}^*\mathrm{Cat}$ as follows:

- weak equivalences are unitary equivalences,
- cofibrations are $*$-functors injective on objects,
- fibrations are $*$-functors such that whenever there is a unitary (equivalence) $u : b \sim Fa'$ for some $a' \in A$ we have $a \in A$ and unitary $v : a \to a'$ such that $Fv = u$, and hence $Fa = b$.

Proof of Theorem. **Axiom 1: Two-out-of-Three law**

Clear.

Axiom 2: Retracts

Consider the diagram in $\mathrm{C}^*\mathrm{Cat}$

$$\begin{array}{ccccc} A & \xrightarrow{P} & A' & \xrightarrow{Q} & A \\ \downarrow f & & \downarrow g & & \downarrow f \\ B & \xrightarrow{I} & B' & \xrightarrow{J} & B \end{array}$$

such that each row is composed to the indentites.

Assuming g is a fibration, we want to show f is a fibration. Let $u : b \sim fa'$ for $b \in B$ and $a' \in A$. Then, $Ib \sim Ifa' = gPa'$. Since g is a fibration, there is $\widetilde{a} \in A'$ with a unitary $\widetilde{a} \to Pa'$ and $g\widetilde{a} = Ib$. Then, $Q\widetilde{a} \sim QPa' = a'$ and $fQ\widetilde{a} = Jg\widetilde{a} = JIb = b$.

Cofibrations are clear.

For weak equivalences, if h is the 'weak inverse' of g and $\widetilde{f} := QhI$, then f and $\widetilde{f}$ show that f is unitary.

Axiom 3: Lifting

Consider

$$\begin{tikzcd} A \arrow[r, "P"] \arrow[d, "\alpha"] & B \arrow[d, "\beta"] \\ C \arrow[r, "Q"] & D \end{tikzcd}$$

Suppose first α is an acylic cofibration and β is a fibration. Take a weak inverse $\widetilde{\alpha} : C \to A$ of α. Since for $c \in C$ we have $Q(c) \sim Q\alpha\widetilde{\alpha}(c) = \beta P\widetilde{\alpha}(c) \in D$, there is $b \in B$ and unitary $u_b : b \to P\widetilde{\alpha}(c)$ and $\beta(b) = Q(c)$. Define $\gamma : C$ as follows. Let $\gamma(c) := b$. For $f : c \to c'$ in C, we define $\gamma(f) : b \to b'$ such that

$$b \sim P\widetilde{\alpha}(c) \xrightarrow{P\widetilde{\alpha}(f)} P\widetilde{\alpha}(c') \sim b'.$$

Now we check the commutation. Since α is an acylic cofibration, by Lemma in the paper, we may assume $\widetilde{\alpha}\alpha = \mathrm{id}_A$. Then, $\beta P(a) = \beta P\widetilde{\alpha}\alpha(a) = \beta P$

$\square$

3 June 4

3.1 Small and large

If $\mathcal{C}$ is a locally small category, then its free cocompletion is a full subcategory of $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Set})$.

Proposition 3.1. *Every representable is continuous.*

Proof. Omitted. $\square$

Proposition 3.2. *A left adjoint functor is cocontinuous and left exact.*

Proof. Omitted. $\square$

Proposition 3.3. *Every presheaf over a small category is a colimit of representables.*

Proof. Let $\mathcal{A}$ be a small category and $X \in \widehat{\mathcal{A}}$. We identify objects of $\mathcal{A}$ and representables to regard $\mathcal{A}$ as a subcategory of $\widehat{\mathcal{A}}$. Then, $X(A)$ is the set of morphisms $A \to X$ for $A \in \mathcal{A}$, and $X(f)$ is the pre-composition with f for $f : A' \to A$. Define a small category $\mathcal{I}$ such that

(i) an object i is given by a pair (A_i, f_i) consisting of $A_i \in \mathcal{A}$ and $f_i : A_i \to X$,

(ii) a morphism $i' \to i$ is given by a morphism $f_{i'i} : A_{i'} \to A_i$ in $\mathcal{A}$ satisfying $f_i f_{i'i} = f_{i'}$.

Then, we have a diagram $\mathcal{I} \to \mathcal{A} : i \mapsto A_i$ because the composition of morphisms in $\mathcal{I}$ is defined by the composition of corresponding morphisms in $\mathcal{A}$. Note that for every $f : A \to X$ with $A \in \mathcal{A}$, there is unique $i \in \mathcal{I}$ such that $f = f_i$, and for every $h : A' \to A$ and $f : A \to X$ with $A, A' \in \mathcal{A}$, there are unique $i, i' \in \mathcal{I}$ such that $f = f_i$ and $f_i h = f_{i'}$. With these i, i', the last condition implies that h induces a morphism $i' \to i$.

We claim $X = \mathrm{colim}_i A_i$ in $\widehat{\mathcal{A}}$. Choose a cocone $g_i : A_i \to Y$ in $\widehat{\mathcal{A}}$ for the diagram A_i. For each $A \in \mathcal{A}$, define a function $g_A : X(A) \to Y(A)$ such that $g_A(f) := g_i$ for $f : A \to X$, where $i \in \mathcal{I}$ is chosen such that $f = f_i$. Then, the family $\{g_A\}$ defines a natural transformation $g : X \to Y$ since for every $h : A' \to A$ in $\mathcal{A}$

$$Y(h)(g_A(f)) = Y(h)(g_i) = g_i h = g_{i'} = g_{A_{i'}}(f_{i'}) = g_{A_{i'}}(f_i h) = g_{A'}(X(h)(f)), \qquad f \in \mathrm{Hom}(A, X),$$

where $i, i' \in \mathcal{I}$ are chosen such that $f = f_i$ and $f_i h = f_{i'}$, because h induces a morphism in $\mathcal{I}$ and g_i is a cocone so that $g_i h = g_{i'}$. Furthermore, we have $g f_i = g_i$ for all $i \in \mathcal{I}$ because for all $A' \in \mathcal{A}$

$$(g f_i)_{A'}(h) = g_{A'}((f_i)_{A'}(h)) = g_{A'}(f_i h) = g_{A'}(f_{i'}) = g_{i'} = g_i h = (g_i)_{A'}(h), \qquad h \in \mathrm{Hom}(A', A_i),$$

where i' is chosen such that $f_i h = f_{i'}$, and the condition $g_{A'}(f_{i'}) = g_{i'}$ forces g is unique, so X satisfies the universal property and the claim follows. $\square$

3.2 Nerves and realizations

Proposition 3.4 (Nerves and realizations)**.** *For a functor $F : \mathcal{S} \to \mathcal{C}$, the* nerve *for F is the restricted Yoneda embedding*

$$N : \mathcal{C} \to \widehat{\mathcal{C}} \xrightarrow{\circ F^{\mathrm{op}}} \widehat{\mathcal{S}},$$

and the realization *for F is the left Kan extension R of F along the Yoneda embedding*

$$\begin{tikzcd} \mathcal{S} \arrow[r, "F"] \arrow[d] \arrow[dr, phantom, "\Downarrow"] & \mathcal{C} \\ \widehat{\mathcal{S}} \arrow[ur, dashed, "R"'] & \end{tikzcd}$$

If $\mathcal{S}$ is essentially small and $\mathcal{C}$ is locally small and cocomplete, then the realization exists and is left adjoint to the nerve.

Proof. For each $S \in \mathcal{S}$ and $C \in \mathcal{C}$, Now let $S \in \widehat{\mathcal{S}}$. Since $\mathcal{S}$ is essentially small, we may write $S = \operatorname{colim}_i S_i$ for a diagram S_i of representables. Since $\mathcal{C}$ is cocomplete, we can define $R(S) := \operatorname{colim}_i F(S_i)$. Then, R is a functor such that for every $F' \in \operatorname{Fun}(\widehat{\mathcal{S}}, \mathcal{C})$ there is a bijection

$$\operatorname{Hom}_{\operatorname{Fun}(\widehat{\mathcal{S}},\mathcal{C})}(R, F') \to \operatorname{Hom}_{\operatorname{Fun}(\mathcal{S},\mathcal{C})}(F, F'Y)$$

by... If $\eta, \eta' : R \to F'$ satisfies $\eta_S = \eta'_S : R(S) \to F'Y(S)$ for all $S \in \mathcal{S}$, then $\eta_S = \eta'_S$ for all $S \in \widehat{\mathcal{S}}$? yes. For $S \in \widehat{\mathcal{S}}$, we can construct a function $R(S) \to F'(S)$.. blahblah

For the adjunction, since we have a bijection

$$\operatorname{Hom}_{\mathcal{C}}(F(S_i), C) = C(F(S_i)) = N(C)(S_i) = \operatorname{Hom}_{\widehat{\mathcal{S}}}(S_i, N(C))$$

by the definition of the nerve and the Yoneda lemma, we can check there is a bijection

$$\operatorname{Hom}_{\mathcal{C}}(R(S), C) = \lim_i \operatorname{Hom}_{\mathcal{C}}(F(S_i), C) = \lim_i \operatorname{Hom}_{\widehat{\mathcal{S}}}(S_i, N(C)) = \operatorname{Hom}_{\widehat{\mathcal{S}}}(S, N(C))$$

by the continuity of representable functors. $\square$

Example 3.5.

(a) $\Delta \to \mathrm{Top}$ gives rise to $|\cdot| \dashv \mathrm{Sing}$.

(b) $\Delta \to \mathrm{Cat}$ gives rise to $\mathrm{h} \dashv N$.

(c) $\Delta \to \mathrm{Set}$ gives rise to π_0 left adjoint to the constant functor.

(d) $\Delta \to \mathrm{Cat}_\Delta$ gives rise to the homotopy coherent nerve $N : \mathrm{Cat}_\Delta \to \mathrm{Set}_\Delta$.

We describe the canonical cosimplicial simplicial category $\Delta \to \mathrm{Cat}_\Delta$. For $[n] \in \Delta$, the corresponding category has $[n]$ as its object set, and the simplicial set of morphisms $\operatorname{Hom}(i, j)$ is given by the nerve of the partially ordered set of strictly increasing paths from i to j in $[n]$, whose order is defined by the subdivision such that $i \to j$ is the largest and $i \to i+1 \to \cdots \to j-1 \to j$ is the smallest. The composition is the concatenation.

For a (combinatorial) model category, the category of fibrant-cofibrant objects is simplicially enriched, and its homotopy coherent nerve is an infinity category.

$$|\cdot| : \mathrm{Set}_\Delta \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} \mathrm{Top} : \mathrm{Sing}$$

4 June 11

4.1 Mapping complexes

Definition (Coproducts and products)**.** Let X and Y be simplicial sets. Since the category of simplicial sets is just another name of the category of presheaves on the simplex category, many categorical constructions that can be done in the category of sets are valid. For example, the initial and terminal simplicial sets are $\varnothing$ and $* := \Delta^0$, and the coproduct and product are given by

$$(X \sqcup Y)_n := X_n \sqcup Y_n, \qquad (X \times Y)_n := X_n \times Y_n.$$

Definition (Mapping complexes)**.** Let X and Y be simplicial sets. The *mapping complex* $\mathrm{Map}(X,Y)$ is a simplicial set defined by

$$\mathrm{Map}(X,Y)_n := \mathrm{Hom}_{\mathrm{Set}_\Delta}(\Delta^n \times X, Y),$$

and the face maps and degeneracy maps are canonically defined by

$$\begin{aligned} d_i^n f_n &:= f_n \circ (\delta_n^i \times \mathrm{id}) \in \mathrm{Hom}_{\mathrm{Set}_\Delta}(\Delta^{n-1} \times X, Y), \\ s_i^n f_n &:= f_n \circ (\sigma_n^i \times \mathrm{id}) \in \mathrm{Hom}_{\mathrm{Set}_\Delta}(\Delta^{n+1} \times X, Y), \end{aligned}$$

for $f_n \in \mathrm{Hom}_{\mathrm{Set}_\Delta}(\Delta^n \times X, Y)$ and $0 \le i \le n$. The simplicial identities for d and s follow from the cosimplicial identities for δ and σ.

Theorem. *The category of simplicial sets* Set_Δ *is cartesian closed.*

Proof. The evaluation map $\mathrm{ev} : \mathrm{Map}(X,Y) \times X \to Y$ between simplicial sets defined by

$$\mathrm{ev}_n : \mathrm{Map}(X,Y)_n \times X_n \to Y_n : (f_n, x_n) \mapsto f_n(\iota_n, x_n),$$

where $\iota_n \in \Delta^n \to \Delta^n$ is the identity, is a simplicial map since

$$\begin{aligned} \mathrm{ev}_n(d_i^n(f_n, x_n)) &= \mathrm{ev}_n(d_i^n f_n, d_i^n x_n) = (d_i^n f_n)_n(\iota_{n-1}, d_i^n x_n) = (f_n)_n(\delta_n^i \iota_{n-1}, d_i^n x_n) = (f_n)_n(\iota_n \delta_n^i, d_i^n x_n) \\ &= (f_n)_n(d_i^n \iota_n, d_i^n x_n) = (f_n)_n(d_i^n(\iota_n, x_n)) = d_i^n (f_n)_n(\iota_n, x_n) = d_i^n(\mathrm{ev}_n(f_n, x_n)) \end{aligned}$$

for $(f_n, x_n) : \Delta^n \to \mathrm{Map}(X,Y) \times X$, and the similar identity for degeneracy maps.

To show the exponential law of mapping complexes, consider the functions of sets

$$\mathrm{Hom}_{\mathrm{Set}_\Delta}(X \times Y, Z) \to \mathrm{Hom}_{\mathrm{Set}_\Delta}(X, \mathrm{Map}(Y,Z)) : f \mapsto f(- \times \mathrm{id}),$$

and

$$\mathrm{Hom}_{\mathrm{Set}_\Delta}(X, \mathrm{Map}(Y,Z)) \to \mathrm{Hom}_{\mathrm{Set}_\Delta}(X \times Y, Z) : g \mapsto \mathrm{ev}(g \times \mathrm{id}).$$

We can check that they are mutually inverses by

$$\mathrm{ev}((f \circ (- \times \mathrm{id})) \times \mathrm{id})(x_n, y_n) = \mathrm{ev}((f \circ (x_n \times \mathrm{id})) \times y_n) = f(x_n \times \mathrm{id})(\iota_n, y_n) = f(x_n, y_n)$$

as $\Delta^n \to Z$ for every $f : X \times Y \to Z$ and $(x_n, y_n) : \Delta^n \to X \times Y$, and

$$(\mathrm{ev}(g \times \mathrm{id}))(- \times \mathrm{id})(x_n) = \mathrm{ev}(g \times \mathrm{id})(x_n \times \mathrm{id}) = \mathrm{ev}(g(x_n) \times \mathrm{id}) = g(x_n)(\iota_n, \mathrm{id}) = g(x_n)$$

as $\Delta^n \times Y \to Z$ for every $g : X \to \mathrm{Map}(Y, Z)$, $x_n : \Delta^n \to X$.

Therefore, the object side has

$$\mathrm{Map}(X \times Y, Z)_n = \mathrm{Hom}_{\mathrm{Set}_\Delta}(\Delta^n \times X \times Y, Z) \cong \mathrm{Hom}_{\mathrm{Set}_\Delta}(\Delta^n \times X, \mathrm{Map}(Y, Z)) = \mathrm{Map}(X, \mathrm{Map}(Y, Z))_n,$$

and the morphism side... □

4.2 Homotopy theory

Definition (Connected components). A *summand* of a simplicial set is a simplicial subset whose degree-wise complement is also a simplicial subset. A simplicial set is called *connected* if it is non-empty and there is no non-trivial summand. A *connected component* of a simplicial set is a connected summand. The set of components of a simplicial set X is denoted by $\pi_0(X)$.

Proposition. *Let X be a simplicial set.*

(a) *π_0 is given by the left Kan extension of the constant functor $\Delta \to \mathrm{Set}$ along the Yoneda embedding.*

(b) *π_0 is exhibited as the coequalizer of the face maps d_0 and d_1 for each simplicial set.*

Proof. (a) As a lemma we first show that for a simplicial map $f : X \to Y$ from connected X there is a unique connected component of Y containing $f(X)$. Let Y' be the smallest summand containing $f(X)$, which exists by taking intersections. Since X and Y' should be non-empty, it is enough to show Y' is connected. If Y'' is a summand of Y', then $f^{-1}(Y'')$ is a summand of X, so $Y'' = Y'$ or $Y'' = \varnothing$, and this says that Y' is connected.

Now we prove the statement. The corresponding nerve functor $\mathrm{Set} \to \mathrm{Set}_\Delta$ of the constant functor $\Delta \to \mathrm{Set}$ is described such that a set S associates a simplicial set $\underline{S}$ such that $\underline{S}_n = S$ as sets for all n and all face and degeneracy maps are identities. It has a canonical bijection between S and $\pi_0(\underline{S})$. Thus, we are enough to show

$$\mathrm{Hom}_{\mathrm{Set}}(\pi_0(X), S) = \mathrm{Hom}_{\mathrm{Set}_\Delta}(X, \underline{S})$$

for any set S, and it easily follows from the special case

$$\mathrm{Hom}_{\mathrm{Set}}(*, S) = \mathrm{Hom}_{\mathrm{Set}_\Delta}(X, \underline{S}),$$

provided X is connected. This is true by the lemma.

(b) Suppose a function $f : X_0 \to S$ between sets satisfies $f d_0 = f d_1$. Since all face and degeneracy maps in the simplicial set $\underline{S}$ are identities, f provides a simplicial map $X \to \underline{S}$. By the adjunction in the part (a), we can induce $\pi_0(X) \to S$, so $\pi_0(X)$ is the coequalizer of d_0 and d_1. □

Definition (Homotopy category). $[X, Y] := \pi_0(\mathrm{Map}(X, Y))$.

Proposition. *Let X be a simplicial set.*

(a) *$\pi_{\leq 1}$ is given by the left Kan extension of the inclusion $\Delta \to \mathrm{Cat}$ along the Yoneda embedding.*

Definition (Homotopy 2-category).

4.3 Kan fibrations and anodynes

Definition (Kan fibrations)**.** A simplicial map $p : Y \to B$ is called a *Kan fibration* if it has the right lifting property with respect to horn inclusions $\Lambda^n_i \to \Delta^n$ for $0 \leq i \leq n$.

Proposition. *Closure properties.*

Definition (Anodynes)**.** Gabriel-Zisman

5

Kan complexes!

5.1 Simplicial homotopy groups

simplicial quotient

Definition 5.1 (Spheres)**.** The *simplicial sphere* of dimension $n \geq 0$ is the simplicial quotient $S^n := \Delta^n/\partial\Delta^n$, which can be canonically understood as a pointed simplicial set. As unpointed simplicial sets, we have $S^n \cong \partial\Delta^{n+1}$ in which sense?

Theorem 5.2. *group structure*

5.2 Whitehead lemma

homotopy equivalence

5.3 Fibrant replacement

5.4 Category of topological spaces

The concrete description of the geometric realization $|S|$ of a simplicial set S is given as follows. topological space $\bigsqcup_{n\geq 0}(S_n \times |\Delta^n|)/\sim$, where the equivalence relation is given by the face and degeneracy maps?

5.1 (Finite simplicial sets)**.**

(a) A simplicial set is finite if and only if it is a compact object of Set_Δ, i.e. the corepresentable functor preserves filtered colimit. 3.6.1.9.

(b) For a simplicial set S, a subset $K \subset |S|$ is compact if and only if it is closed and there is a finite simplicial subset $S' \subset S$ such that $K \subset |S'|$. In particular, $|S|$ is compact if and only if S is finite. 3.6.1.11.

A left adjoint generally does not preserve finite limits, which is equivalent to the exactness.